B. Writtin Analysis

• Why does a linked-list stack push at the head instead of the tail?

A linked-list stack pushes at the head because it makes adding and removing elements faster. The head represents the top of the stack, so both operations can be done in constant time, or \(O(1)\). If we push at the tail without maintaining extra information, we may need to traverse the list. Therefore, using the head is more efficient.

• Why must an array queue be circular?

An array queue uses a circular structure so it can reuse empty positions created when elements are removed. Without a circular structure, the rear pointer may reach the end of the array even when there are empty spaces at the beginning. Using the modulo operator allows the rear pointer to return to the beginning when needed.